

IOT Based Landslide Detection in Hilly Area

Kishor B. Bhargale

Electronics and Telecommunication Pimpri
Chinchwad College of Engineering and
Research
Pune, India.
kishor.bhargale@pccoer.in

Aditya Mane

Electronics and Telecommunication Pimpri
Chinchwad College of Engineering and
Research,
Pune, India
adimane282000@gmail.com

Yatish Patil

Electronics and Telecommunication Pimpri
Chinchwad College of Engineering and
Research
Pune, India.
patilyatish592000@gmail.com

Rahul G. Mapari

Electronics and Telecommunication Pimpri
Chinchwad College of Engineering and
Research
Pune, India.
rahul.mapari@pccoer.in

Harshal Bhosale

Electronics and Telecommunication Pimpri
Chinchwad College of Engineering and
Research
Pune, India.
bhosaleharshal5445@gmail.com

Abstract— As per the records more than 5000 people are buried alive and there are more than 4 billion USD in losses every year due to landslide disasters globally. In India, the problem is quite severe, as India happens to be the landslide capital of the world where on average about thousand people lose their life every year on account of landslides. The problem with the landslide sensing technologies existing is that these technologies are costly technologies like conventional geotechnical monitoring. Thus, this article presents the early landslide detection based on various parameters of the hilly area such as moisture, rain level, vibrations and accelerometer sensor outputs. The proposed system provides an alert buzzer to the proximity using Fuzzy logic Controller. The anticipated scheme is implemented for the real time scenario and data is stored on the cloud for the future reference. It provides a simple, feasible and low-cost landslide monitoring and warning system.

Keywords: *IoT, Wireless Sensor Network, Landslide, Low-cost, Real-time Monitoring, Thing-Speak.*

I. INTRODUCTION

Landslides are a major factor we are currently facing. According to records, about 32% of the earth's land is covered by forests. But deforestation is on the rise to make human life more comparable, and this leads to landslide-like problems [1][2]. Landslides are caused by natural activities and other human activities. Natural activities include heavy rainfall and ice deposits, and the rapid melting of ice causes water to mix with the soil mass on the slope, making it more likely that the land will slide [3]. Or human activities include planting, deforestation, machinery vibration. Most of the landslides are on the lower slopes of the highlands. The greater the movement and vibration in the rock, the greater the rate of landslides [4][5][6].

Landslides are a very common problem in India in the hilly region and there are very few technologies which are available to help the common people [7][8]. Recently, Internet of Things has attracted the attention of the researchers because of its ability to control the things remotely, to connect the millions of devices together, ability to access real time data, support for the cloud computing and artificial intelligence. It has been widely used for many applications such as security applications, home automation, building automation, transportation, logistics, industrial automation, robotics and control, agriculture field monitoring, etc [9][10][11][12].

After seeing this major problem we have developed a unique technology which can not only monitor the landslide but also provide early warning to the general public with the feature of Real-time monitoring using the cloud with proper graphs and alerts. As this landslide monitoring system is very low-cost product this could be easily developed and can be deployed at several locations considering its low cost.

Remaining article is organized as follows: Section 2 gives brief information about previous work on Landslide Detection System. Section 3 gives brief information on the Proposed Methodology of the work. Section 4 gives detailed information on the Hardware Setup, Flowchart, Results and Discussions, Flow Chart and Block Diagram of the proposed system. Section 5 gives an information on Conclusion and References.

II. RELEVANCE/BACKGROUND

To capture the issues of controlling the life loss associated with this disaster, there are lot of landslide monitoring measures by which the early warning systems are developed. When we looked into the literature we found that these existing methods are range from microscopic to macroscopic. The landslide can be monitored by collecting images from satellites which is based on remote sensing technology [13] and collect information from the satellites and doing proper algorithms by which we can understand the deformation of the soil on the earth surface with the help of satellite images [14]. The other way comes the micro sensing where the installation of proper sensors in the landslide prone locations and according to the site locations we deploy the type of sensors such as piezometers which can be used to monitor the moment of happening of movement in the particular locations [15], other than this we also have methods associated with GPS [16] or UAV [17] methods by which we see what is the relative distance of the relative moment between two fixed points by using GPS methods and a lot of other methods and these are have been successful over the time but what we found was majorly the three problems associated with this particular systems.

The most important factor when talking about any technology is the cost of the system which is very high.

Technologies which we have designed is 1000 times less the cost of a conventional geotechnical monitoring system [18] because of this low cost we can deploy thousands of systems at so many landslides prone locations across India and the world using this technology

The second issue is with deployment so by taking an example if suppose I want to install one battery sensor at the hill top such as piezometer, it is very difficult for carrying up the instruments or equipment associated with the deployment.

Coming to the main problem when we talk about the “real-time monitoring” it's very difficult to monitor and maintain data at a particular time for such a mass piece of land. As it doesn't support real-time monitoring so thereby it cannot give you timely warnings which is a major fallback for previous technologies.

III. PROPOSED METHODOLOGY

This article mainly focuses on the low-cost development and easy to use for detection of landslide and for monitoring system. This System consists of sensor part, Controller at the landslide side and from there the data will be received on the Cloud with the help of IOT. To cover the more area, several sensor nodes are used and all are integrated with one common main controller. The Proposed system uses soil moisture sensor, accelerometer sensor, Vibration Sensor, Rain sensor. From all this sensor we sense the physical data and send it to the pre-processing Block called as Data Acquisition and Amplification. From Sensor Part We are collecting the all the parameter which required for detection of the landslide such as moisture, rain density, slop, displacement. From this Data we are good Enough for detecting the landslide. This data sanded to the amplification block and into that the real time data is converting into the digital data and this digital data is send to the controller. With the help on controller and esp8266 Wi-Fi module data is send to the cloud that is to the Think Speak. The process of proposed system is described in Fig. 1.

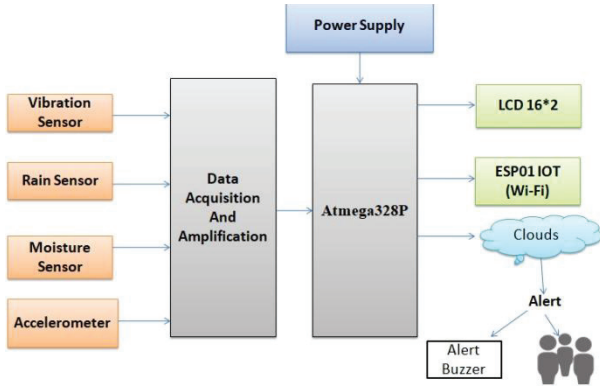


Fig. 1 Block diagram of proposed system

Here, the ATMEGA328p controller is used which is a 8-bit micro-controller based on RISK architecture. The motive to use Atmega328p Controller is it is open source, easy to available and low-cost and also work on low power. Atmega328p controller is programmed with the help of Arduino software IDE. In this project for programming embedded C Language is used. With the help of Controller and ESP8266 Wi-fi Module the data is sent on the cloud. The proposed system uses The Thing-Speak Cloud. On the cloud with the help of MATLAB which is a powerful tool it is easy to use and present the data in the form of graph and that can be easily readable too. And through the analysis and observing the data we are send the alert to the people.

Basically, the landslide is causes due to excess rainfall and due to rapidly change the slop. So, the are moral focus

on moisture data, rain density and vibration in the land. Based on these values we are categories the alert system into three parts as No Alert, Alert and High Alert. In No Alert if the graph of the all the sensor is stable or below the Fixed value which set to the middle alert then there is no any Landslide possibility so in this case, the alert message is not send to the people. For Alert if any one of the sensor values has exceed the standard value or spick of the graph is rapidly changes then we are sending the alert to the citizen that be alert as there might be chances of landslide. If the two or more-sensor value is exceed the standard value or if spike of the any two sensor is changes as rapidly then the High Alert is send to the citizen that be alert there is more chances of landslide be prepared for relocation and also leave the place immediately. And last alert that very danger is send when the all the sensor data value is exceeding the standard value or all the sensor's graph spikes is higher then it sends the alert in the red very Danger alert to the citizen as well as to all the landslide monitoring station as immediately leave the places as there is landslide is occurs. During the very danger alert the alert message also send to the rescue team for emergency .In General Landslide causes due to excess rainfall and due to rapid change in the slope and movement in rocks. So, to overcome this problem the proposed system is Landslide Detection and Monitoring System. In this system we are categories the System into Four part as the sensor Part, Data Amplification Part, Controller part and output part. In sensor Part We are Using the Rain Sensor, Soil moisture sensor, Vibration Sensor, Accelerometer sensor. The Sensor Part plays the role for collecting the physical data for detecting the landslide such as Displacement, Moisture, slop and rain density. Once the data is collected from the sensor part then it sends the data to the Data Acquisition and Amplification part to process the data which is convert the physical data into the digital data by sampling the signal and again it send to the controller block. Controller blockreceive the digital data and with the help of ESP01 IOT Wi- fi module is send measured data to the cloud and also to the LCD display for cross verification. Here we are using the LCD16*2 and Atmega328p micro-controller. For data collection we are using the Thing-Speak Cloud and finally based on the condition the alert message is send into three alert messages as No Alert, Alert and High Alert to the people.

Given below is the block diagram for proposed fuzzy logic-based Landslide Detection System. So in Fig No-2 the designed model in which the four parameters are Vibration, Moisture, Rain and Accelerometer which are sent to the Mamdani model and the Landslide output is received as No Alert, Alert and High Alert shown in the Fig. 2.

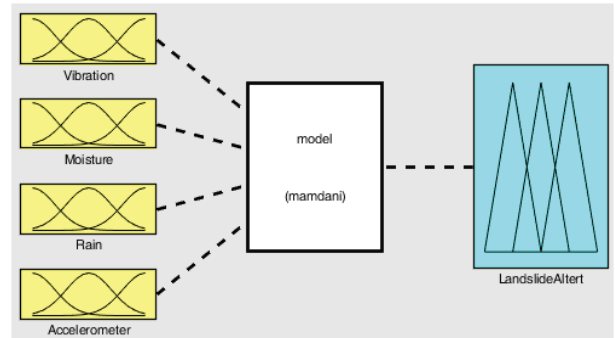


Fig. 2 Proposed fuzzy logic-based landslide detection system

Various Levels of Input and Output variables used for controlling Landslide Alert System are given in Table 1:

TABLE I. RANGE OF VARIOUS SENSORS

Variable Type	Sensor	Levels	Range
Input Variable	Vibration	Low(L)	0-300
		Medium(M)	250-700
		High(H)	550-1000
	Moisture	Low(L)	0-300
		Medium(M)	250-600
		High(H)	550-950
	Rain	Low(L)	0-2.25
		Medium(M)	1.75-2.75
		High(H)	2.25-5
Output	Buzzer	Low(L)	2-6g
		Medium(M)	4-12g
		High(H)	10-16g
		No Alert	0-55s
		Alert	45-260s
		High Alert	250-300s

Fig. 3 provides the details about membership function for vibration sensor input where low range is defined in between 0-300, the medium range is 250-700 and the high range is 550-1000. The membership function equations for the vibration sensor are given in equation 1-3.

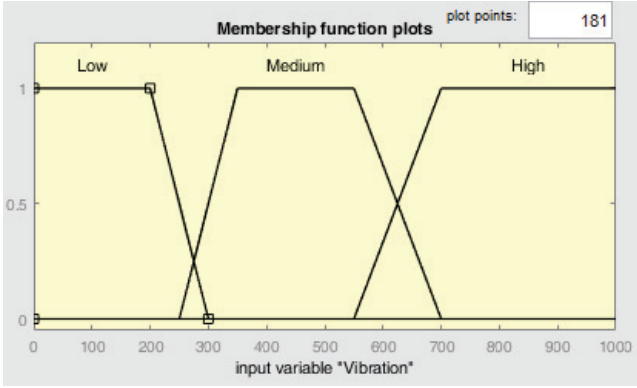


Fig. 3 Membership function for vibration sensor

$$Vibration_{Low} = \begin{cases} 1 & 0 \leq Vb \leq 200 \\ \frac{300-x}{100} & 200 \leq Vb \leq 300 \end{cases} \quad (1)$$

$$Vibration_{Med} = \begin{cases} \frac{x-250}{100} & 350 \leq Vb \leq 550 \\ \frac{700-x}{150} & 250 \leq Vb \leq 350 \\ 1 & 550 \leq Vb \leq 700 \end{cases} \quad (2)$$

$$Vibration_{High} = \begin{cases} \frac{x-550}{150} & 700 \leq Vb \leq 1000 \\ 1 & 550 \leq Vb \leq 700 \end{cases} \quad (3)$$

The fuzzy membership function for moisture sensor is shown in Fig. 4 and its equivalent equations are described in equation 4-6.

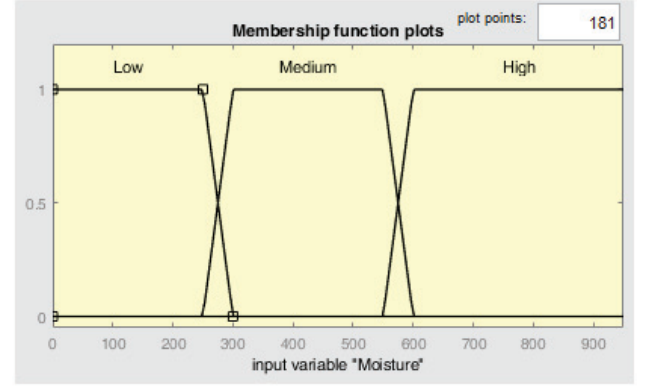


Fig. 4 Membership function for Moisture sensor

$$Moisture_{Low} = \begin{cases} 1 & 0 < Mo < 300 \\ \frac{300-x}{50} & 250 < Mo < 300 \end{cases} \quad (4)$$

$$Moisture_{Med} = \begin{cases} \frac{x-250}{50} & 250 \leq Mo \leq 300 \\ \frac{50}{600-x} & 600 \leq Mo \leq 950 \end{cases} \quad (5)$$

$$Moisture_{High} = \begin{cases} \frac{x-550}{50} & 550 < Mo < 600 \\ 1 & 600 < Mo < 950 \end{cases} \quad (6)$$

The fuzzy membership function for rain sensor is shown in

Fig. 5 and its equivalent equations are described in equation 7-9.

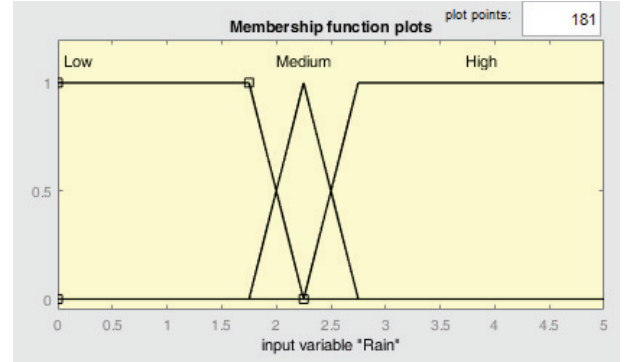


Fig. 5 Membership function for Rain sensor

$$Rain_{Low} = \begin{cases} 1 & 0 \leq x \leq 1.75 \\ \frac{2.25-x}{0.50} & 1.75 \leq x \leq 2.25 \end{cases} \quad (7)$$

$$Rain_{Med} = \begin{cases} \frac{x-1.75}{0.50} & 1.75 \leq x \leq 2.25 \\ \frac{2.75-x}{0.50} & 2.25 \leq x \leq 2.75 \end{cases} \quad (8)$$

$$Rain_{High} = \begin{cases} \frac{x-2.25}{0.50} & 2.25 \leq x \leq 2.75 \\ 1 & 2.75 \leq x \leq 5 \end{cases} \quad (9)$$

The fuzzy membership function for Accelerometer sensor is shown in Fig. 6 and its equivalent equations are described inequation 10-12.

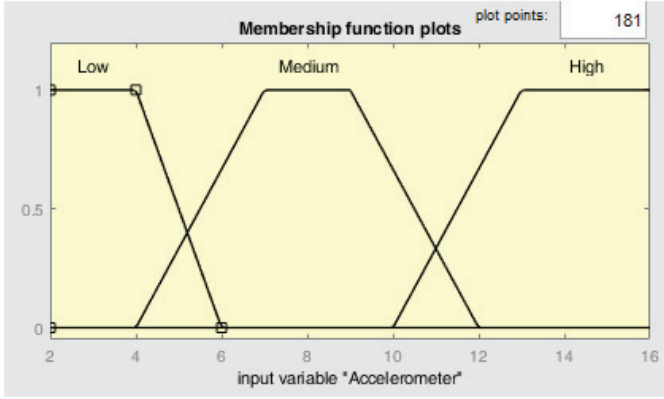


Fig. 6 Membership function for Accelerometer

$$Accelerometer_{Low} = \begin{cases} 1 & 2 \leq Ac \leq 4 \\ 6-x & 4 \leq Ac \leq 6 \end{cases} \quad (10)$$

$$Accelerometer_{Med} = \begin{cases} \frac{x-4}{2} & 4 \leq Ac \leq 7 \\ \frac{12-x}{3} & 7 \leq Ac \leq 9 \\ 1 & 9 \leq Ac \leq 12 \end{cases} \quad (11)$$

$$Accelerometer_{High} = \begin{cases} \frac{x-10}{3} & 10 \leq Ac \leq 13 \\ 1 & 13 \leq Ac \leq 16 \end{cases} \quad (12)$$

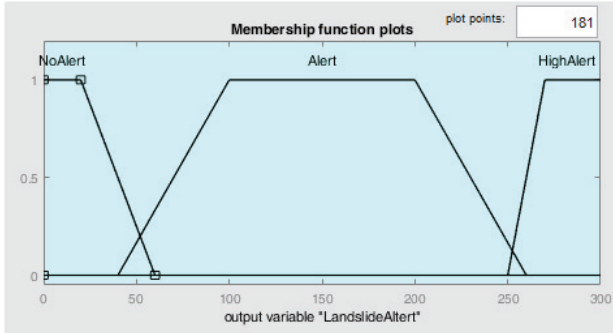


Fig 7. Membership function for Landslide Alert

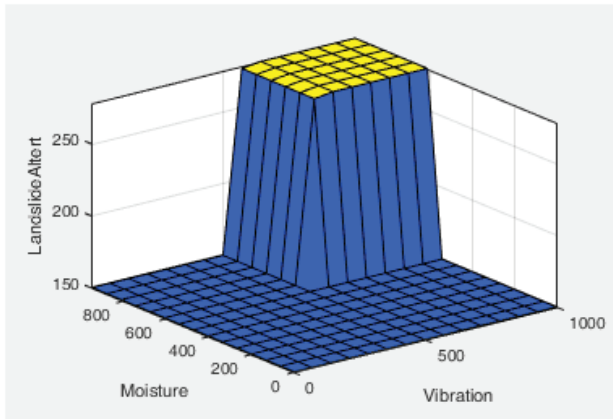


Fig 8. Surface plot for Moisture vs Vibration vs LandslideAlert

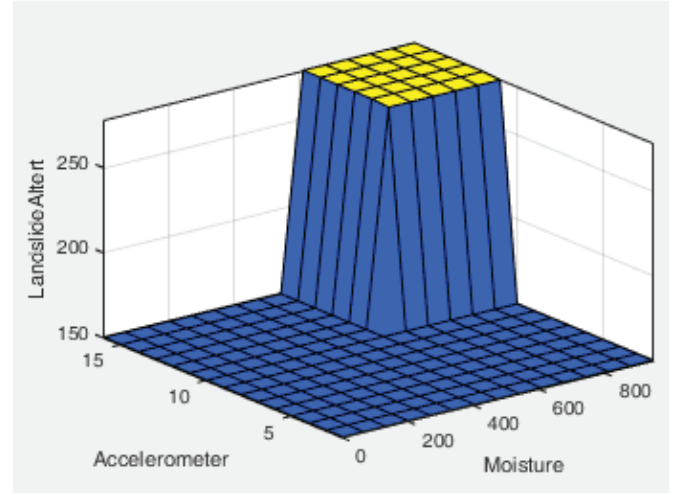


Fig 9.-Surface plot for Moisture vs Accelerometer vs Landslide Alert

The above Fig. 8 represents the data between Moisture and Vibration Sensor input on which the Landslide Alert is determined and the Fig. 9 determines the surface plot between Accelerometer and Moisture Sensor.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Hardware Setup

The details about various sensors and the component used for development of proposed system are given in Table 2:

TABLE II. HARDWARE COMPONENT SPECIFICATION

Hardware Component	Module	Specification
Vibration Sensor	SW-420	This sensor used to measure the change within acceleration, pressure, temperature, strain by changing to an electrical charge
Moisture Sensor	RC-A-4079	It is used to sense the moisture in the soil.
Rain Sensor		It is a device which is used to detect therainfall.
Accelerometer	ADXL 335	Accelerometer sensors are ICs thatmeasure acceleration, which is the changein speed per unit time

The rule for the fuzzy logic controllers are described in Table 3.

TABLE III. OUTPUT OF BUZZER

Input Variables				Output Variables
Vibration	Rain	Moisture	Accelerometer	Buzzer
Low	Low	Low	Low	No Alert
High	Low	Low	Medium	Alert
High	Low	Medium	Low	Alert
High	Medium	Low	Low	Alert
Low	High	High	Low	High Alert
Low	High	Medium	Medium	High Alert
High	High	High	High	High Alert
Medium	Low	Low	Low	No Alert
Low	Low	Medium	Low	No Alert
Medium	Medium	Medium	Medium	Alert

B. Flow-Chart

The flow chart of the proposed system is illustrated in Fig. 10.

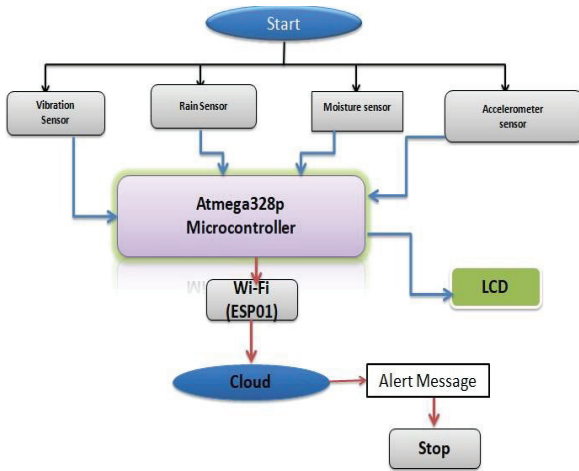


Fig. 10 Flow chart of the proposed scheme

C. Results and Discussions

The different control scenarios for the no alert, alert and high alert are shown in Fig. 11-13.



Fig. 11 No Alert scenario

For the fuzzy system the Rule Viewers for “No Alert Scenario” will be Vibration=538, Moisture=197, Rain =1.4, Accelerometer = 3.62 and Landslide Alert = 20.8 seconds.



Fig. 12 Alert scenario

For the fuzzy system the Rule Viewers for “Alert Scenario” will be Vibration=750, Moisture=431, Rain =2.97, Accelerometer = 3.62 and Landslide Alert = 150 seconds.

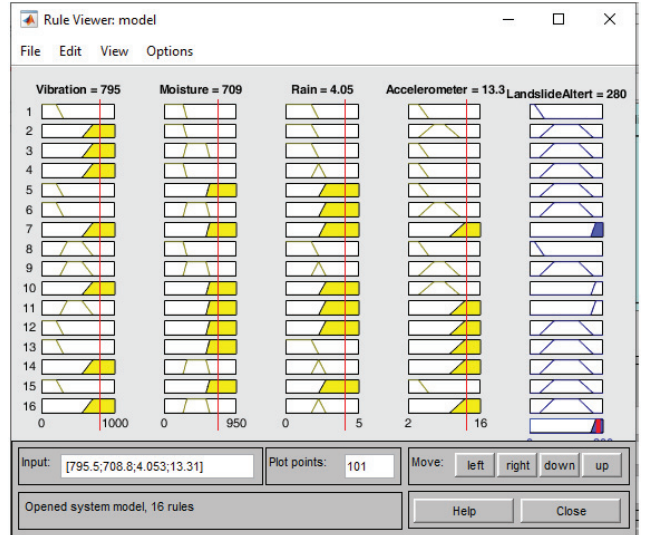


Fig. 13 High Alert scenario

For the fuzzy system the Rule Viewers for “High Alert Scenario” is shown in Fig. 13 with the Vibration=795, Moisture=709, Rain =4.05, Accelerometer = 13.3 that produces Landslide Alert = 280 seconds.

The data collected using different sensor is stored on the cloud for the future reference using Things-Speak cloud platform.

V. CONCLUSION

Thus, this article presents the automatic landslide detection based on the moisture, vibration, rain and accelerometer properties of the hilly region. It provides the fuzzy logic based control of the alter system to alert the people about landslide calamity. This system helps to early detection of landslide detection so that proper preventive or care should be taken to avoid the social and economic causalities. This System it easy to monitor and also easy to detect the landslide in the hilly region. The System is capable of sensing the data 24*7 hrs. and sending it properly on the

cloud and also, we can monitor it at any time any from anywhere. In future, the performance of the system can be evaluated for the real time scenario with wireless sensor network and internet of things.

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